

Introduction to programmable logic devices

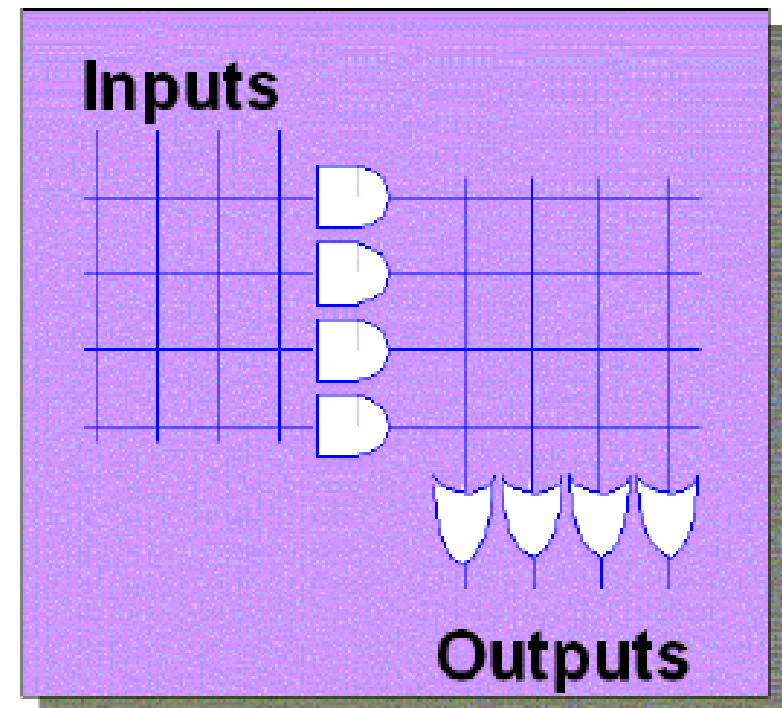
History of programmable logic devices

SPLD Architectures

PLA

- Two programmable planes
- Any Combination of ANDs / ORs
- Sharing of AND terms across multiple OR's
- Highest logic density available to user
- High Fuse count, Slower than PALs
- Programmable Logic Array - PLA

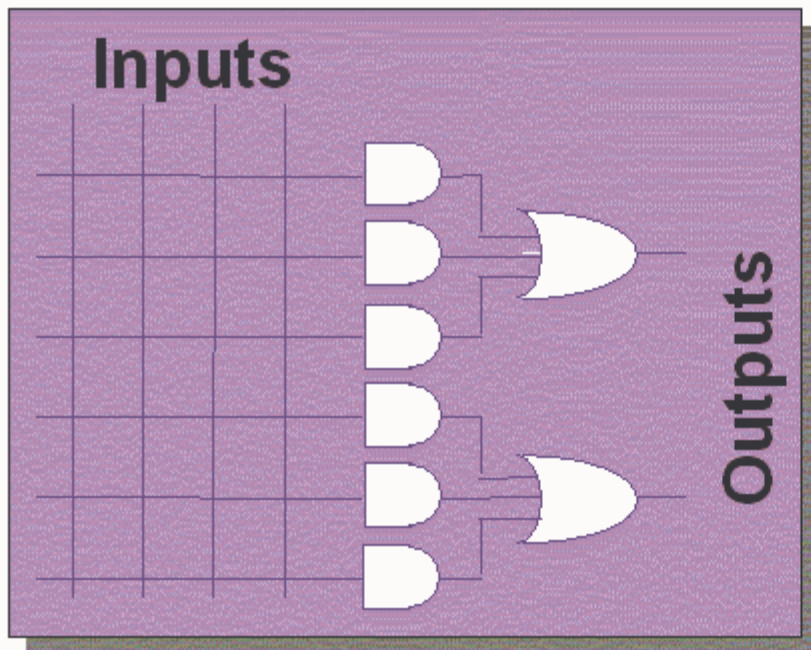
Signetics - Cline - 1975



SPLD Architectures

PAL

MMI - Birkner - 1978



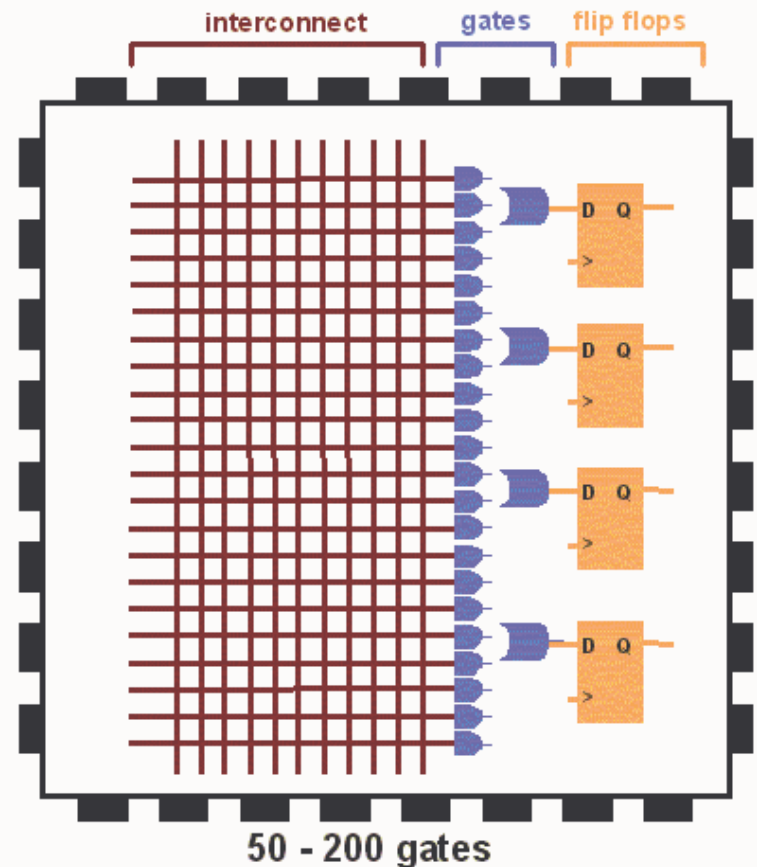
- One programmable plane - AND / Fixed OR
- Finite combination of ANDs / ORs
- Medium logic density available to user
- Lower Fuse count, Faster than PLAs (at this time fabricated on a 10um process)
- Programmable Array Logic - PAL

Complex Programmable Logic Devices (CPLDs)

CPLD Architecture

- Central, Global Interconnect
- Simple, Deterministic Timing
- Easily routed
- PLD Tools add only interconnect
- Wide, fast complex gating

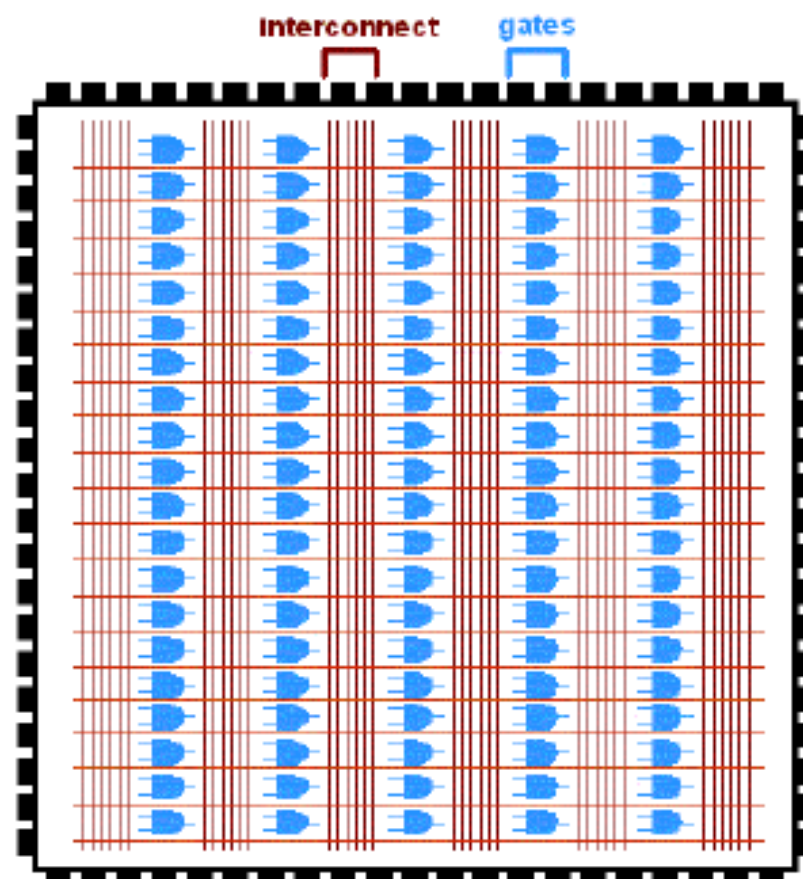
Design can run upto 200MHz



FPGA Architecture

- Channel Based Routing
- Post Layout Timing
- Tools More Complex than CPLDs
- Fine Grained
- Fast register pipelining

"What if we could develop the equivalent of a circuit board full of standard logic parts (like TTL & PAL devices) on a single high density programmable logic chip?"



1,000,000+ gates

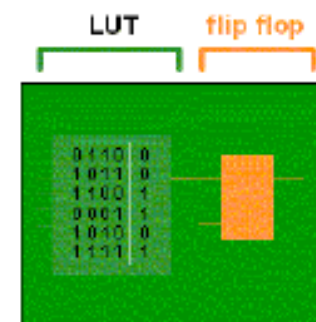
Xilinx - Freeman - 1985

Digital Logic History

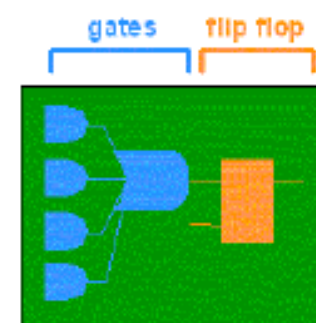
FPGA - Field Programmable Gate Array

2 types of *FPGAs*

- Reprogrammable (SRAM-based)
 - SRAM determines interconnect
 - SRAM defines logic in Look Up Table (LUT)
- One-time Programmable (OTP)
 - Interconnect is anti-fuse
 - Logic is traditional gates



SRAM logic cell



OTP logic cell

FPGA vendors

- Two major vendors
 - Xilinx
 - Altera

Xilinx Spartan-3E (3rd series FPGAs)

Table 1: Summary of Spartan-3E FPGA Attributes

Device	System Gates	Equivalent Logic Cells	CLB Array (One CLB = Four Slices)				Distributed RAM bits ⁽¹⁾	Block RAM bits ⁽¹⁾	Dedicated Multipliers	DCMs
			Rows	Columns	Total CLBs	Total Slices				
XC3S100E	100K	2,160	22	16	240	960	15K	72K	4	2
XC3S250E	250K	5,508	34	26	612	2,448	38K	216K	12	4
XC3S500E	500K	10,476	46	34	1,164	4,656	73K	360K	20	4
XC3S1200E	1200K	19,512	60	46	2,168	8,672	136K	504K	28	8
XC3S1600E	1600K	33,192	76	58	3,688	14,752	231K	648K	36	8

Xilinx Spartan-6 (6th series FPGAs)

Spartan-6 LX FPGAs

Optimized for Lowest-Cost Logic, DSP, and Memory
(1.2V, 1.0V)

Part Number	XC6SLX4	XC6SLX9	XC6SLX16	XC6SLX25	XC6SLX45	XC6SLX75	XC6SLX100	XC6SLX150
Slices ⁽¹⁾	600	1,430	2,278	3,758	6,822	11,662	15,822	23,038
Logic Cells ⁽²⁾	3,840	9,152	14,579	24,051	43,661	74,637	101,261	147,443
CLB Flip-Flops	4,800	11,440	18,224	30,064	54,576	93,296	126,576	184,304
Maximum Distributed RAM (Kb)	75	90	136	229	401	692	976	1,355
Block RAM (18 Kb each)	12	32	32	52	116	172	268	268
Total Block RAM (Kb) ⁽³⁾	216	576	576	936	2,088	3,096	4,824	4,824
Clock Management Tiles (CMT) ⁽⁴⁾	2	2	2	2	4	6	6	6
Maximum Single-Ended Pins	132	200	232	266	358	408	480	576
Maximum Differential Pairs	66	100	116	133	179	204	240	288
DSP48A1 Slices ⁽⁵⁾	8	16	32	38	58	132	180	180
Endpoint Block for PCI Express®	—	—	—	—	—	—	—	—
Memory Controller Blocks	0	2	2	2	2	4	4	4
GTP Low-Power Transceivers	—	—	—	—	—	—	—	—
Commercial ⁽¹⁰⁾	-1L, -2, -3	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N
Industrial ⁽¹⁰⁾	-1L, -2, -3	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N	-1L, -2, -3, -3N
Configuration Memory (Mb)	2.7	2.7	3.7	6.4	11.9	19.6	26.5	33.8

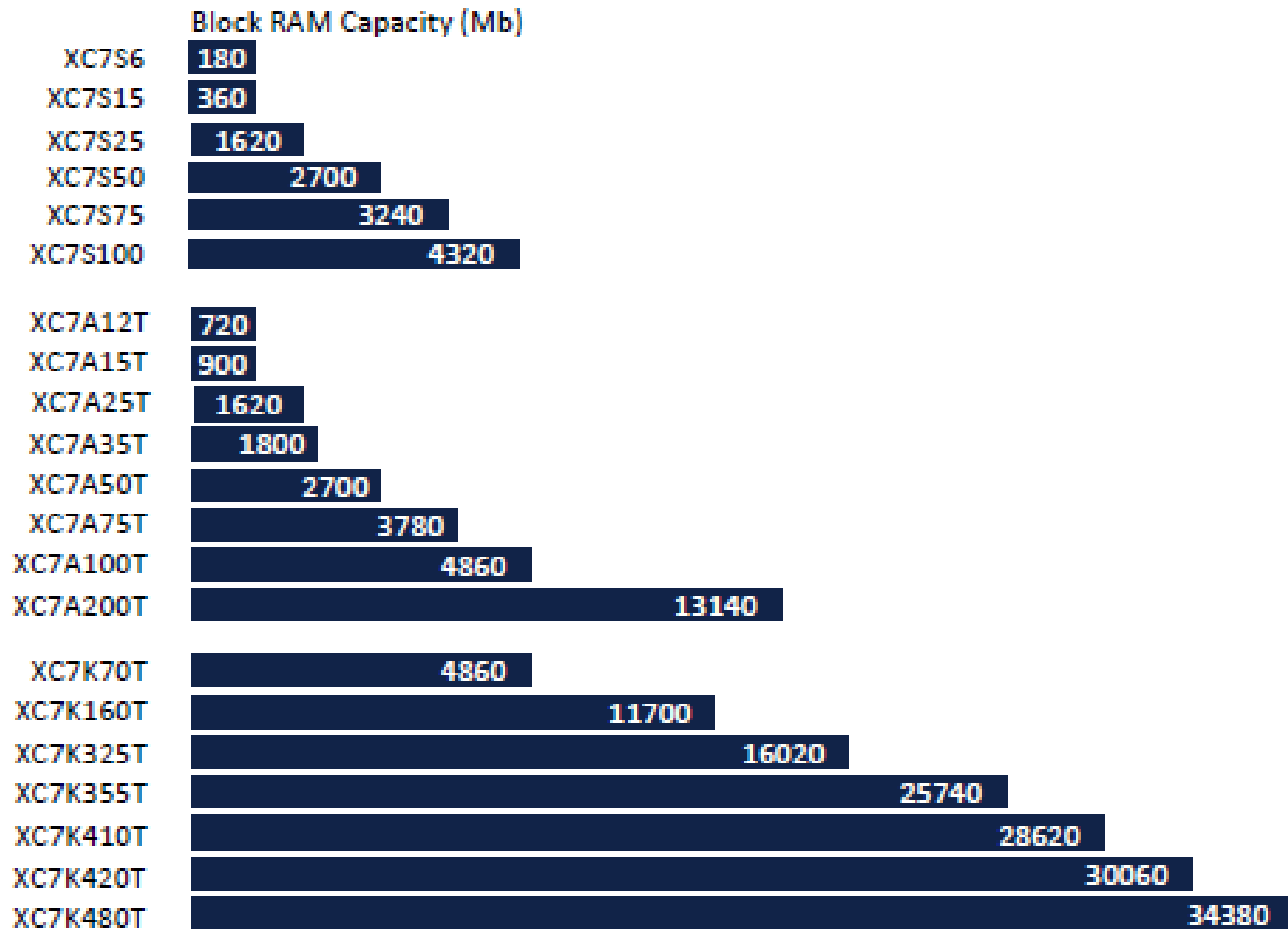
Xilinx Artix-7 (7th series FPGAs)

Transceiver Optimization at the Lowest Cost and Highest DSP Bandwidth
(1.0V, 0.95V, 0.9V)

Part Number	XC7A12T	XC7A15T	XC7A25T	XC7A35T	XC7A50T	XC7A75T	XC7A100T	XC7A200T
Logic Cells	12,800	16,640	23,360	33,280	52,160	75,520	101,440	215,360
Slices	2,000	2,600	3,650	5,200	8,150	11,800	15,850	33,650
CLB Flip-Flops	16,000	20,800	29,200	41,600	65,200	94,400	126,800	269,200
Maximum Distributed RAM (Kb)	171	200	313	400	600	892	1,188	2,888
Block RAM/FIFO w/ ECC (36 Kb each)	20	25	45	50	75	105	135	365
Total Block RAM (Kb)	720	900	1,620	1,800	2,700	3,780	4,860	13,140
CMTs (1 MMCM + 1 PLL)	3	5	3	5	5	6	6	10
Maximum Single-Ended I/O	150	250	150	250	250	300	300	500
Maximum Differential I/O Pairs	72	120	72	120	120	144	144	240
DSP Slices	40	45	80	90	120	180	240	740
PCIe® Gen2 ⁽¹⁾	1	1	1	1	1	1	1	1
Analog Mixed Signal (AMS) / XADC	1	1	1	1	1	1	1	1
Configuration AES / HMAC Blocks	1	1	1	1	1	1	1	1
GTP Transceivers (6.6 Gb/s Max Rate) ⁽²⁾	2	4	4	4	4	8	8	16

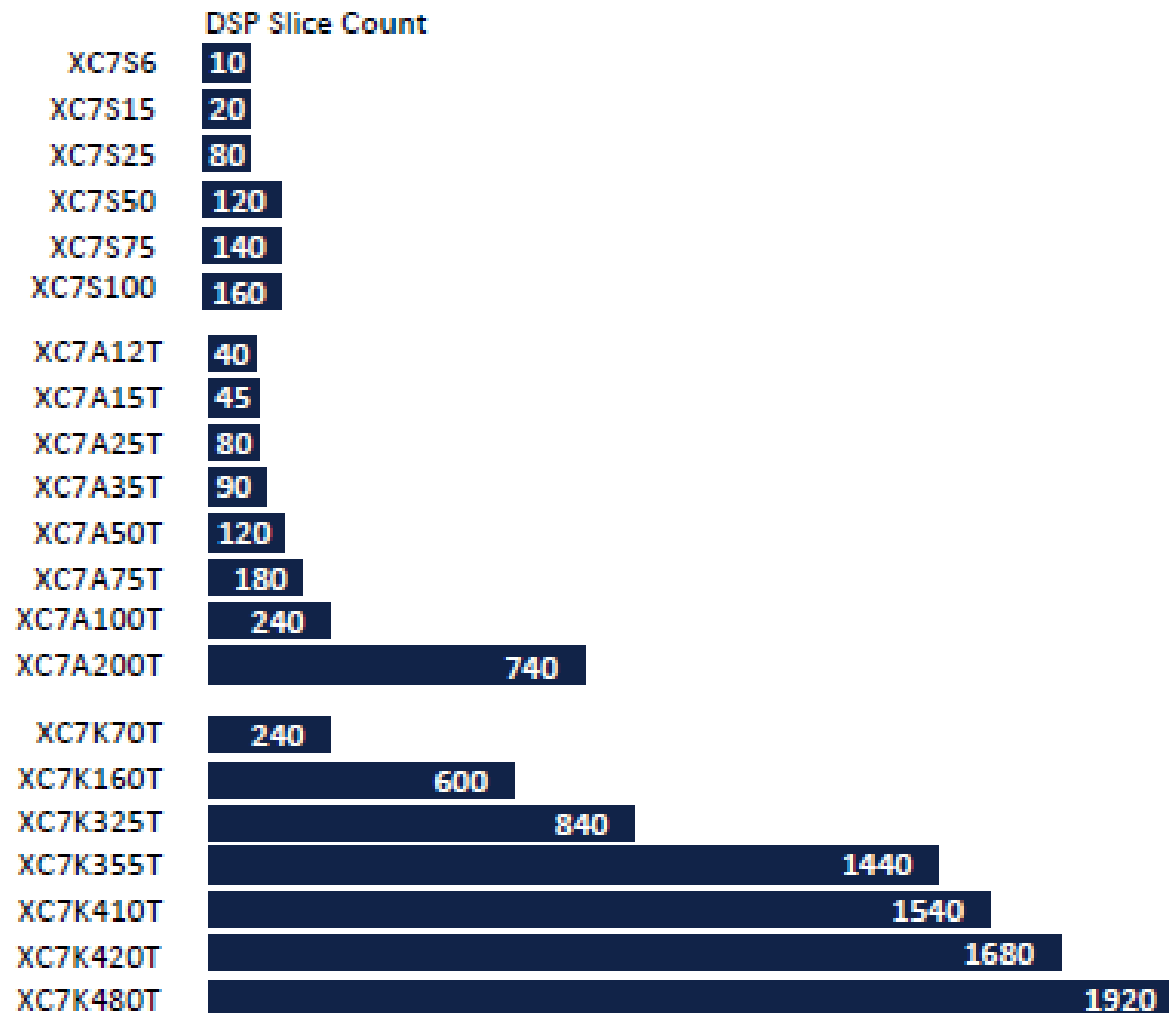
(7th series FPGAs)

Block RAM Metrics



(7th series FPGAs)

Digital Signal Processing Metrics



Spartan-II^E Architecture

Delay Locked Loop (DLL)

Configurable Logic Block (CLB)

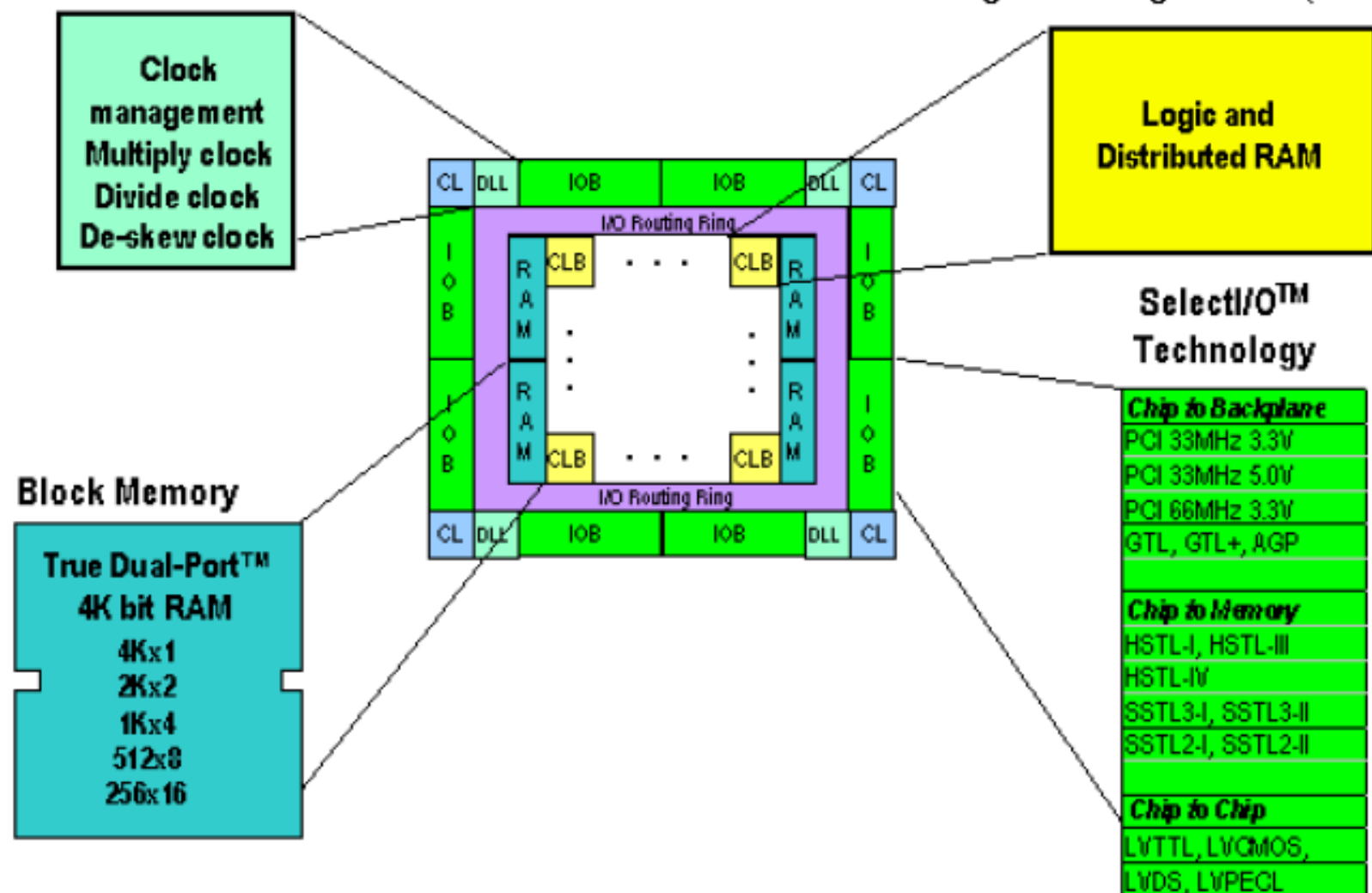
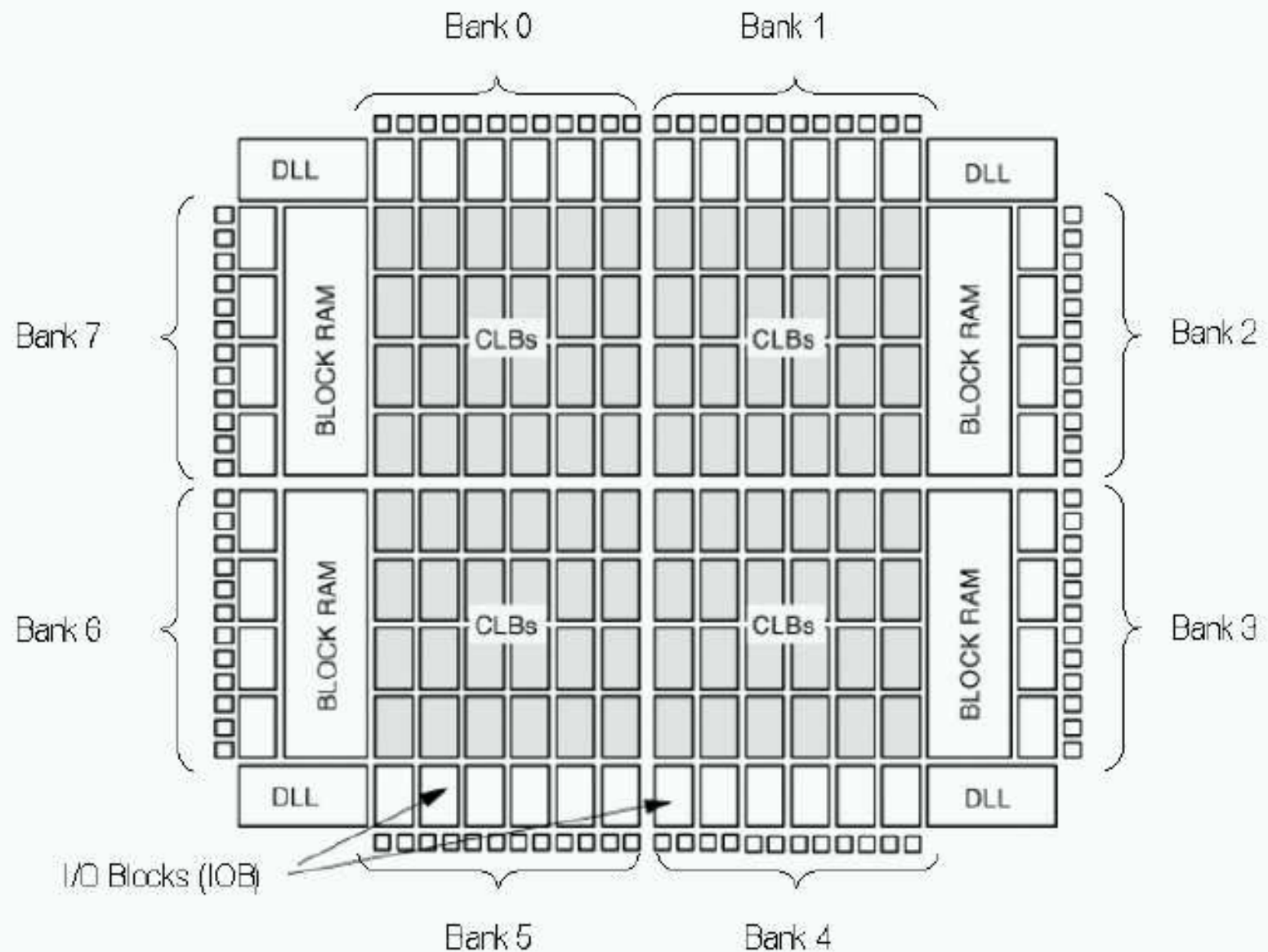
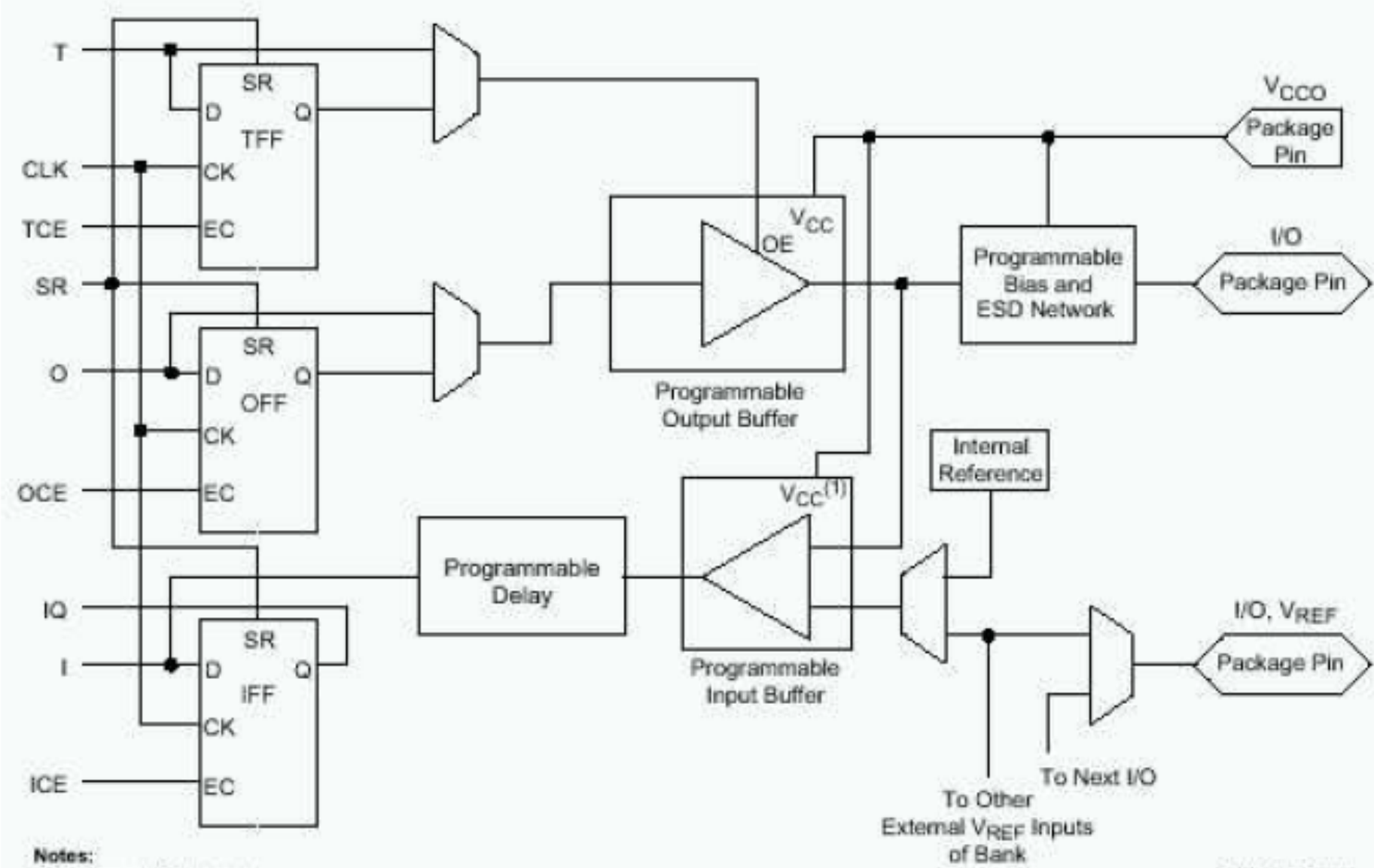


Figure 2.5 Spartan IIE Architecture

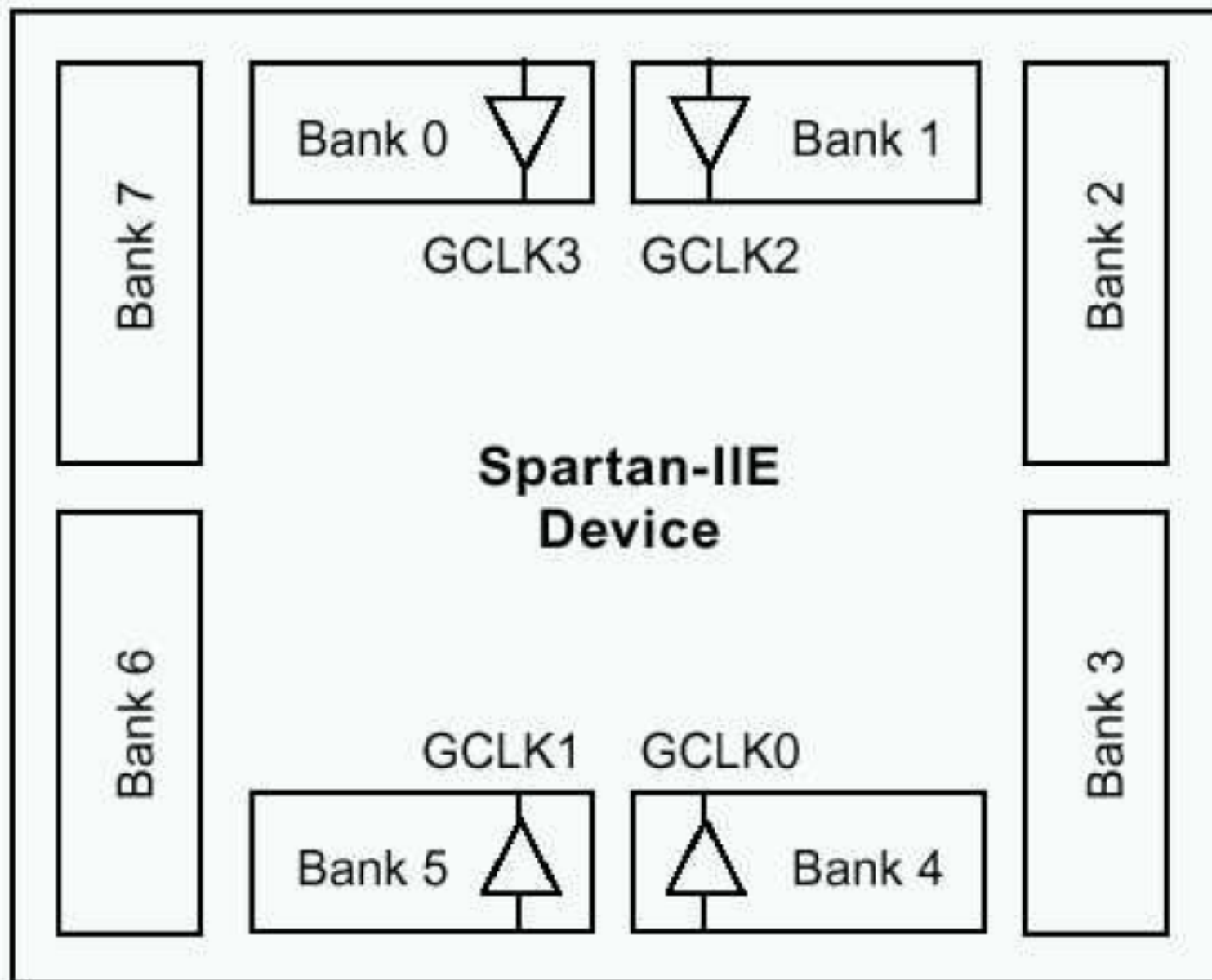
Spartan-IIe Block Diagram

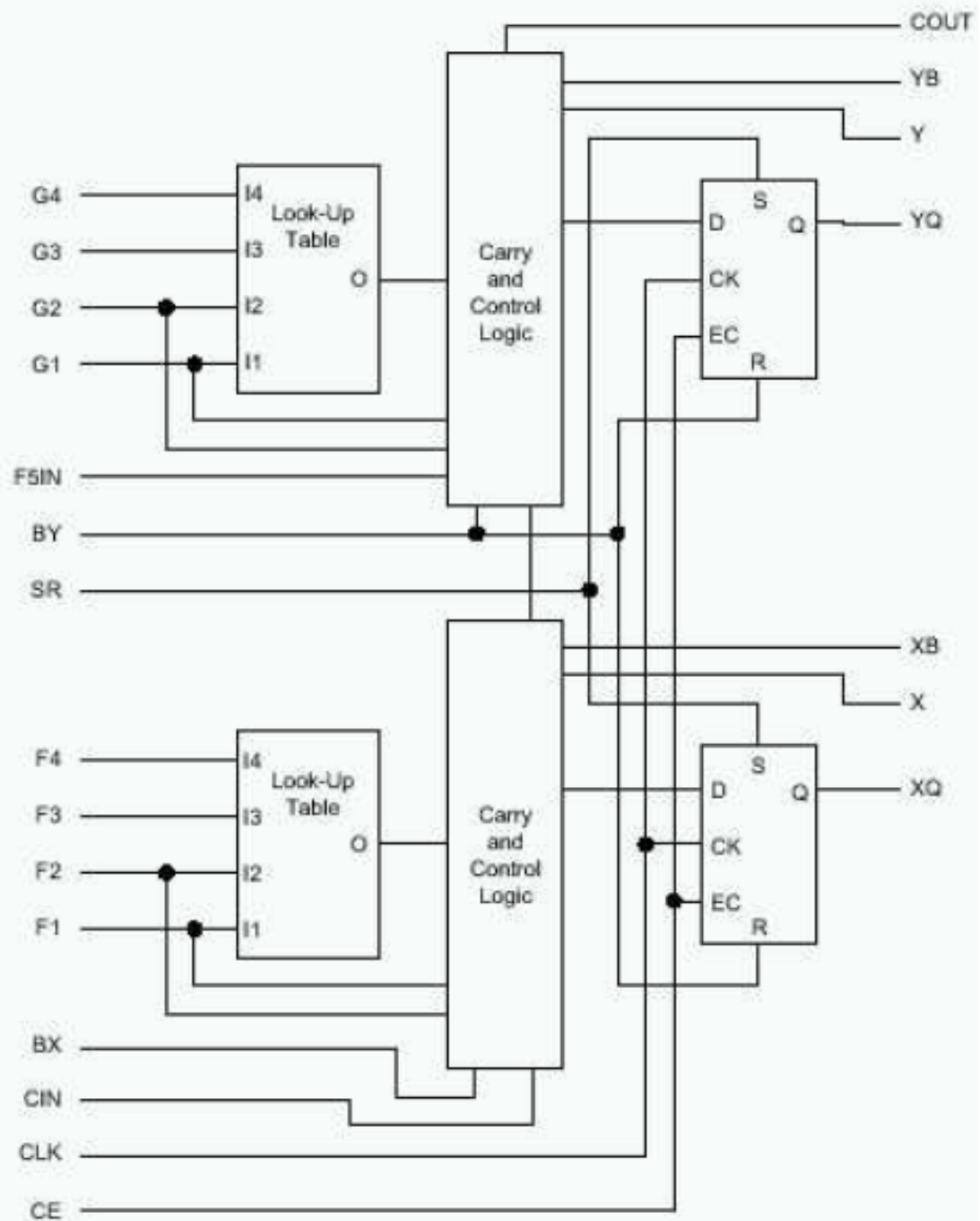


Spartan-II E Input/Output Block



Spartan-II^E Banking of I/O Standards





Spartan IIE Logic Cell

I/O capabilities

Spartan-3E FPGAs support the following single-ended standards:

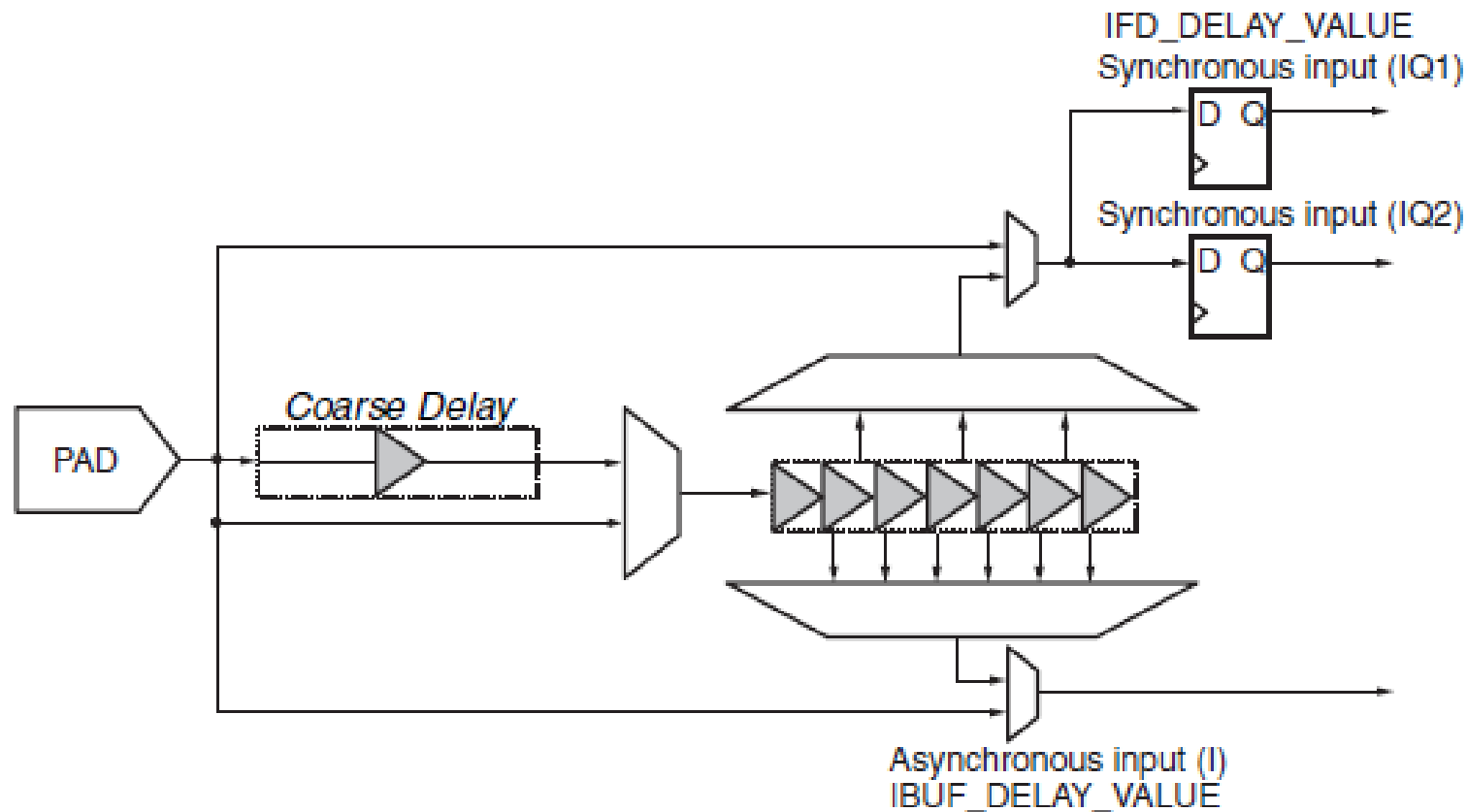
- 3.3V low-voltage TTL (LVTTTL)
- Low-voltage CMOS (LVCMOS) at 3.3V, 2.5V, 1.8V, 1.5V, or 1.2V
- 3V PCI at 33 MHz, and in some devices, [66 MHz](#)
- HSTL I and III at 1.8V, commonly used in memory applications
- SSTL I at 1.8V and 2.5V, commonly used for memory applications

I/O capabilities

Spartan-3E FPGAs support the following differential standards:

- LVDS
- Bus LVDS
- mini-LVDS
- RSDS
- Differential HSTL (1.8V, Types I and III)
- Differential SSTL (2.5V and 1.8V, Type I)
- 2.5V LVPECL inputs

Input delay functions



UG331_c10_09_011508

Figure 6: Programmable Fixed Input Delay Elements

I/O block

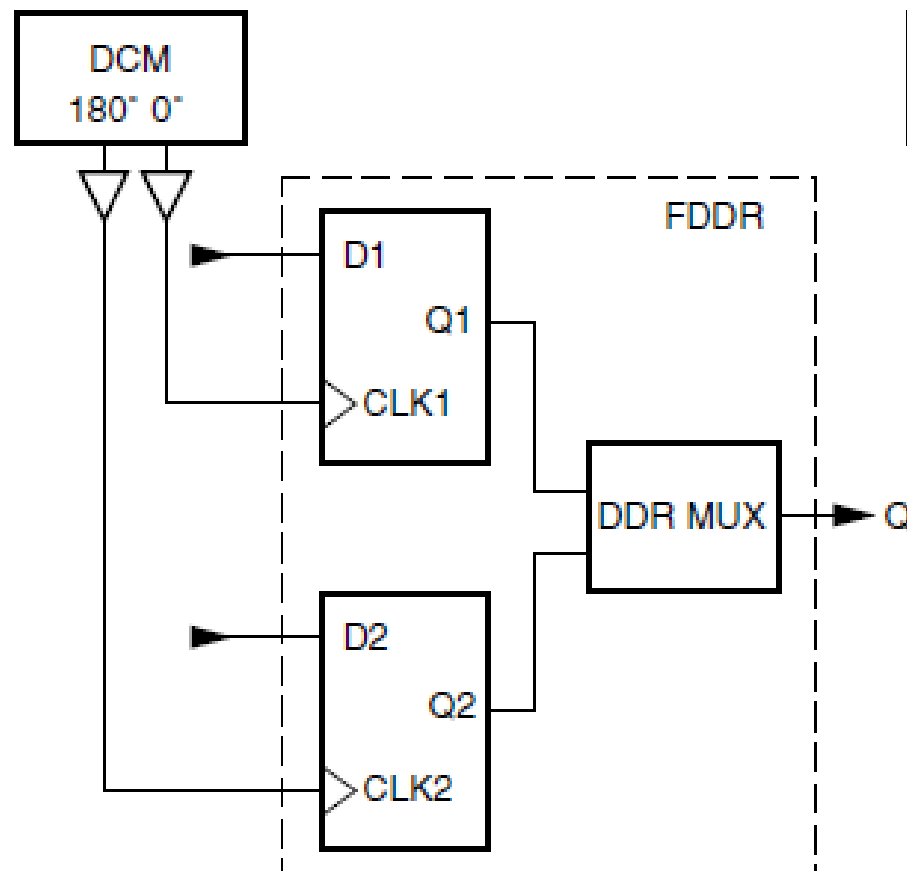
Storage Element Functions

There are three pairs of storage elements in each IOB, one pair for each of the three paths. It is possible to configure each of these storage elements as an edge-triggered D-type flip-flop (FD) or a level-sensitive latch (LD).

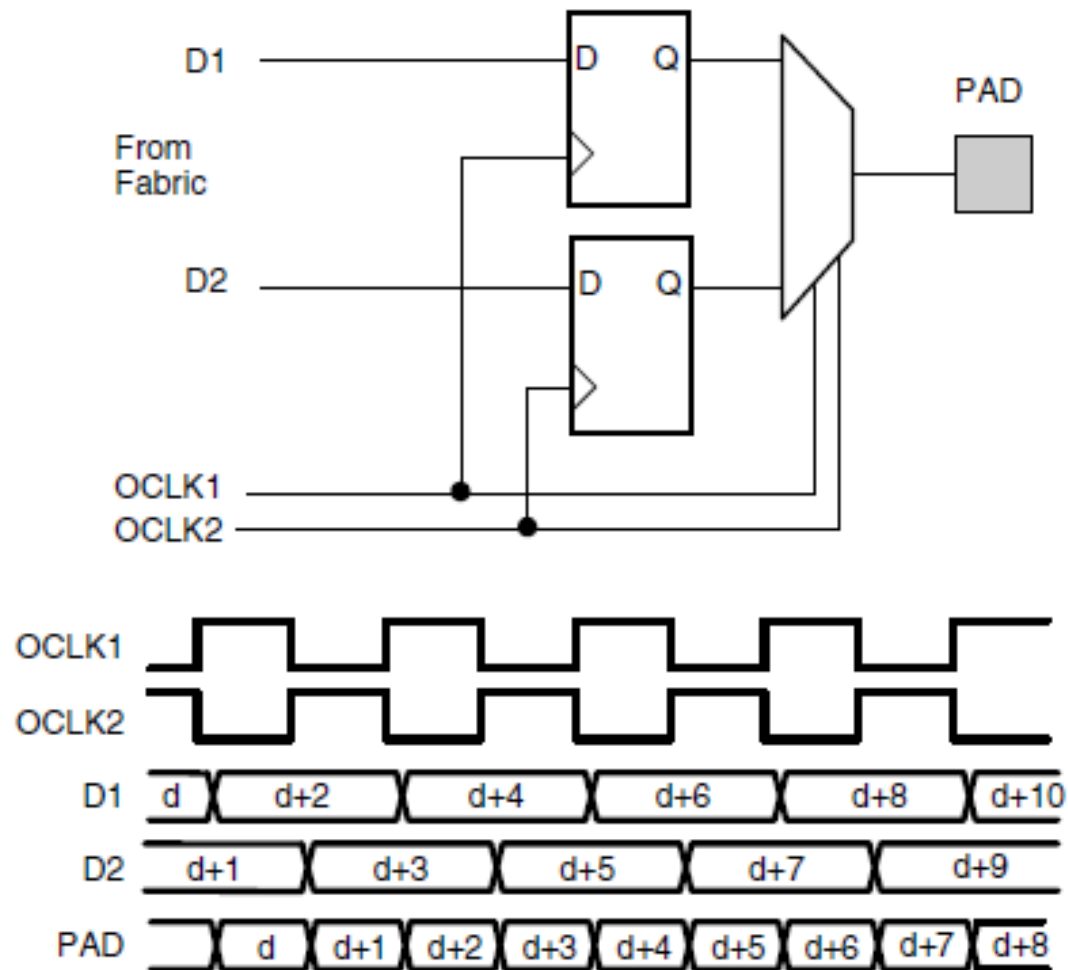
The storage-element pair on either the Output path or the Three-State path can be used together with a special multiplexer to produce Double-Data-Rate (DDR) transmission.

Double-Data-Rate Transmission

Double-Data-Rate (DDR) transmission describes the technique of synchronizing signals to both the rising and falling edges of the clock signal. Spartan-3E devices use register pairs in all three IOB paths to perform DDR operations.



Output DDR



DS312-2_23_030105

Figure 10: Output DDR

Input DDR

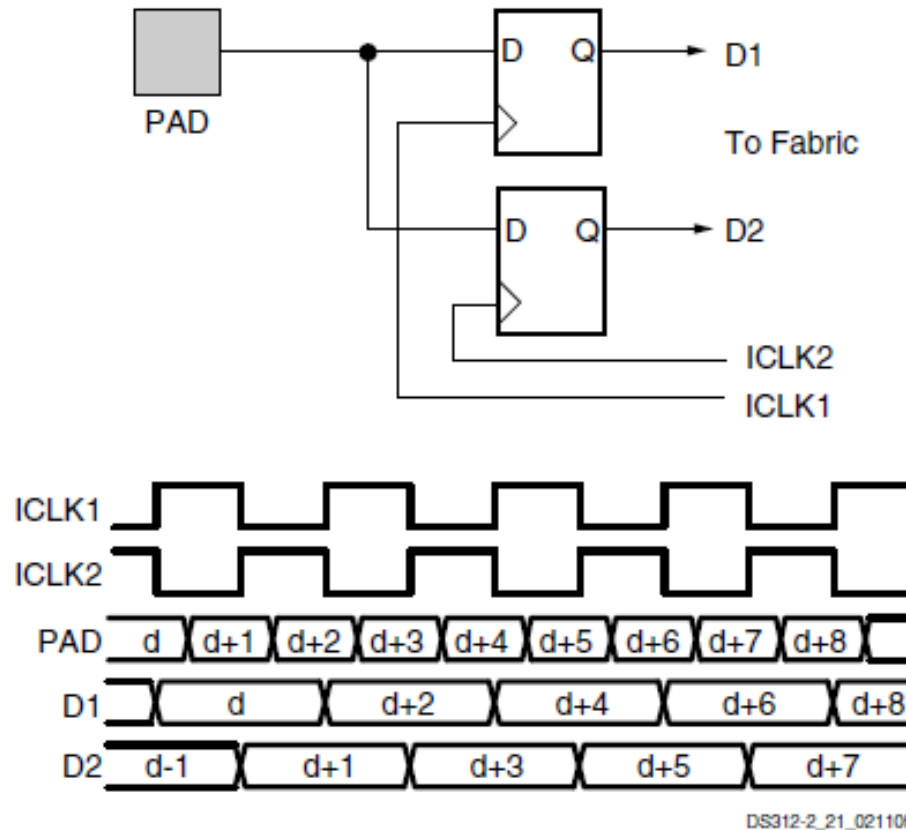
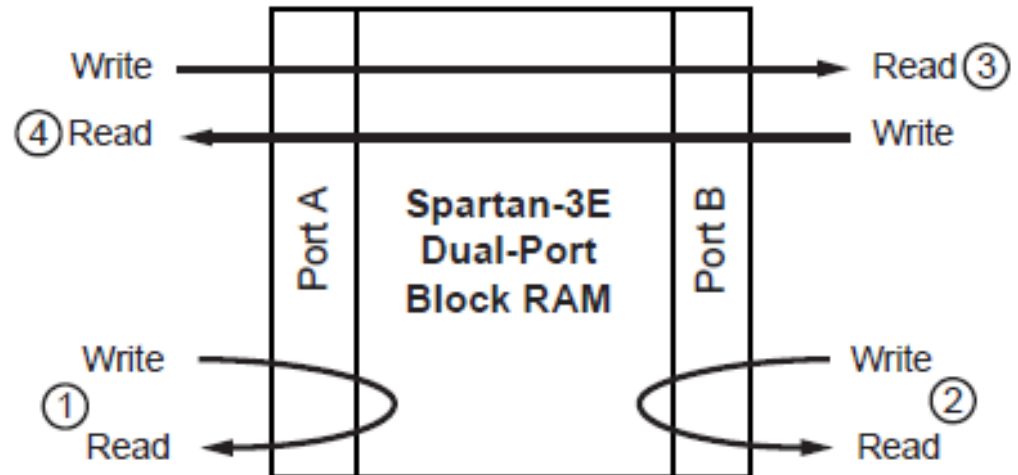


Figure 8: Input DDR (without Cascade Feature)

BLOCKRAM

- One block equals 18Kbits (36Kb in 7th series)
- Supports dual port structure
- Four data paths

1. Write to and read from Port A
2. Write to and read from Port B
3. Data transfer from Port A to Port B
4. Data transfer from Port B to Port A



DS312-2_01_020705

Figure 30: Block RAM Data Paths

Block RAM modes of operation

- Write first
- Read first
- No-change

Write first

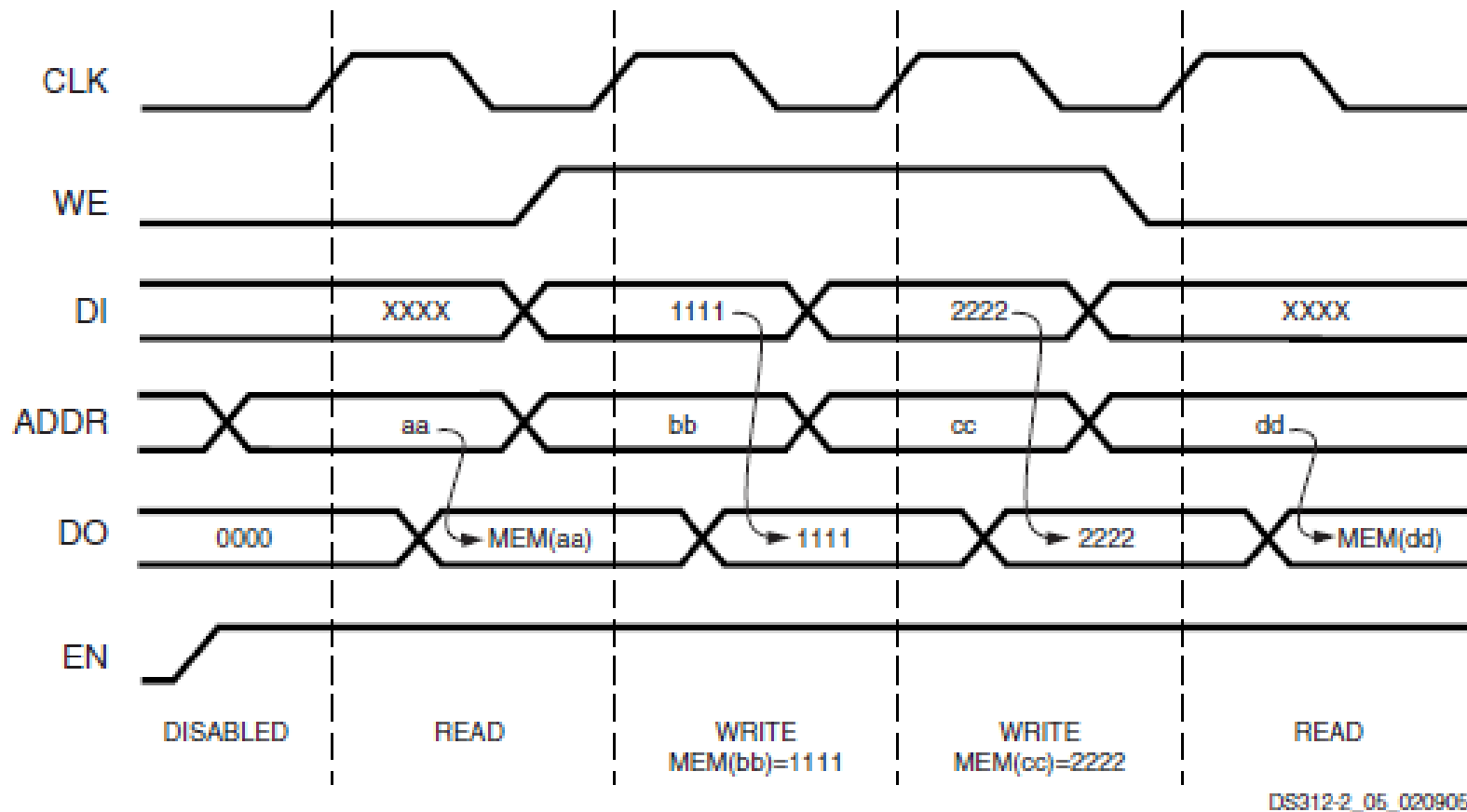
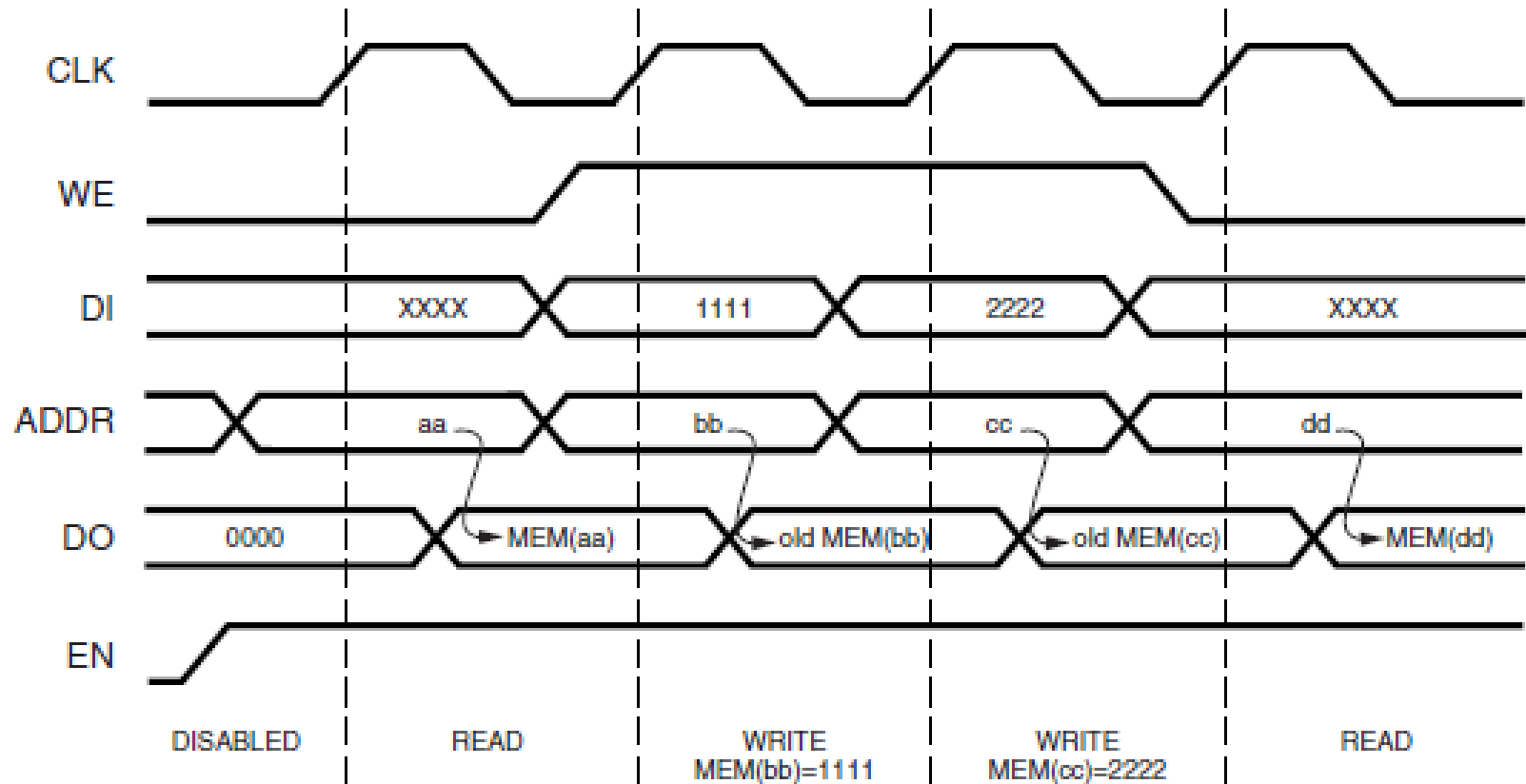


Figure 33: Waveforms of Block RAM Data Operations with WRITE_FIRST Selected

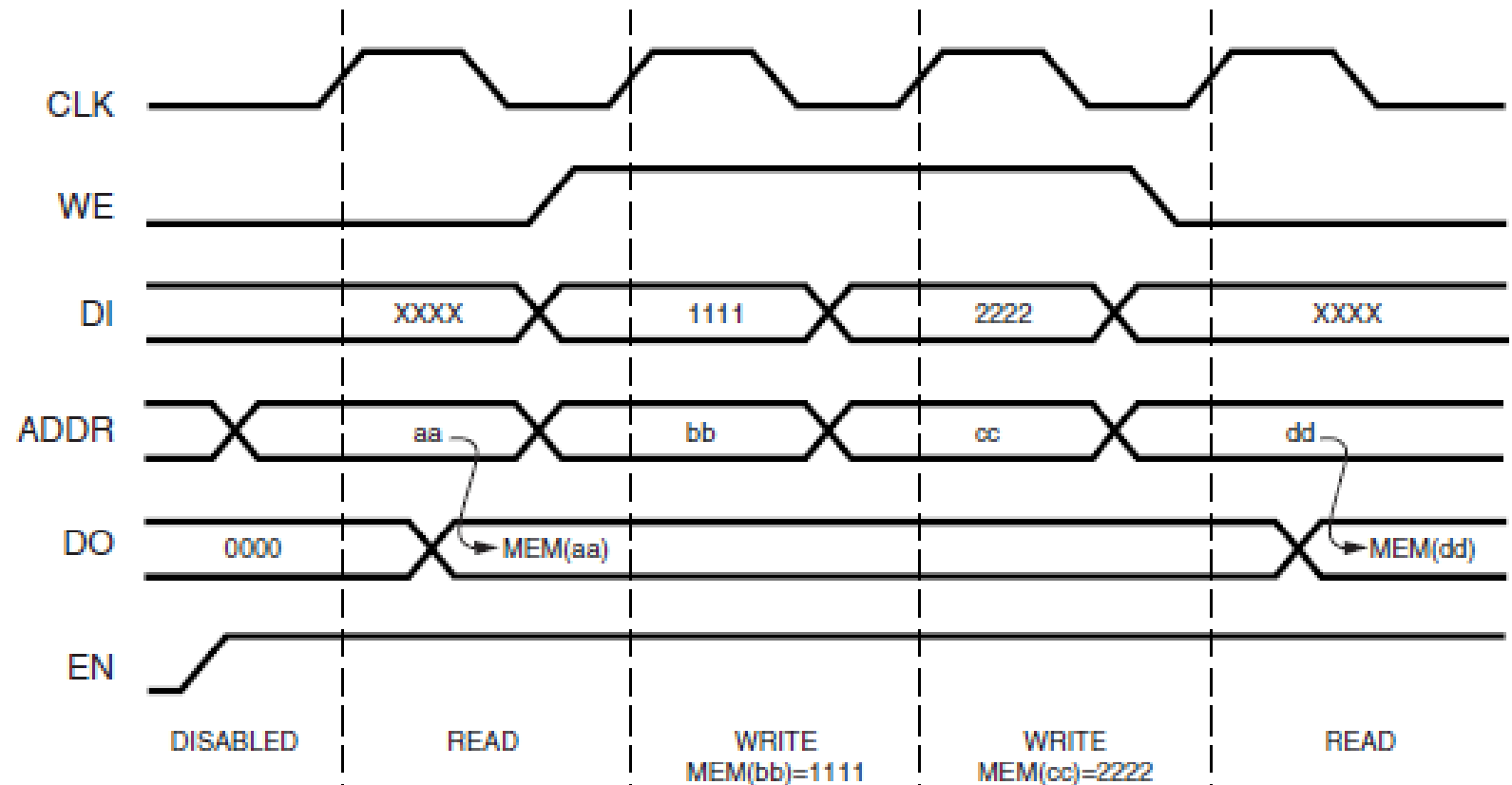
Read first



DS312-2_06_020805

Figure 34: Waveforms of Block RAM Data Operations with READ_FIRST Selected

No change



DS312-2_07_020905

Figure 35: Waveforms of Block RAM Data Operations with NO_CHANGE Selected

Read/write modes

- No-change

```
always @(posedge <clock>)
  if (<ram_enable>)
    if (<write_enable>)
      <ram_name>[<address>] <= <input_data>;
    else
      <output_data> <= <ram_name>[<address>];
```

- Read first

```
always @(posedge <clock>)
  if (<ram_enable>) begin
    if (<write_enable>)
      <ram_name>[<address>] <= <input_data>;
    <output_data> <= <ram_name>[<address>];
  end
```

- Write first

```
always @(posedge <clock>)
  if (<ram_enable>) begin
    if (<write_enable>) begin
      <ram_name>[<address>] <= <input_data>;
      <output_data> <= <input_data>;
    end
    else
      <output_data> <= <ram_name>[<address>];
  end
```

Dual port BLOCKRAM

```
parameter RAM_WIDTH = <ram_width>;
parameter RAM_ADDR_BITS = <ram_addr_bits>;

(* RAM_STYLE="{AUTO | BLOCK | BLOCK_POWER1 | BLOCK_POWER2}" *)
reg [RAM_WIDTH-1:0] <ram_name> [(2**RAM_ADDR_BITS)-1:0];
reg [RAM_WIDTH-1:0] <output_dataA>, <output_dataB>;

<reg_or_wire> [RAM_ADDR_BITS-1:0] <addressA>, <addressB>;
<reg_or_wire> [RAM_WIDTH-1:0] <input_dataA>;

always @(posedge <clockA>)
    if (<enableA>) begin
        if (<write_enableA>)
            <ram_name>[<addressA>] <= <input_dataA>;
        <output_dataA> <= <ram_name>[<addressA>];
    end

always @(posedge <clockB>)
    if (<enableB>) begin
        if (<write_enableB>)
            <ram_name>[<addressB>] <= <input_dataB>;
        <output_dataB> <= <ram_name>[<addressB>];
    end
```

Write and read at different clock rate

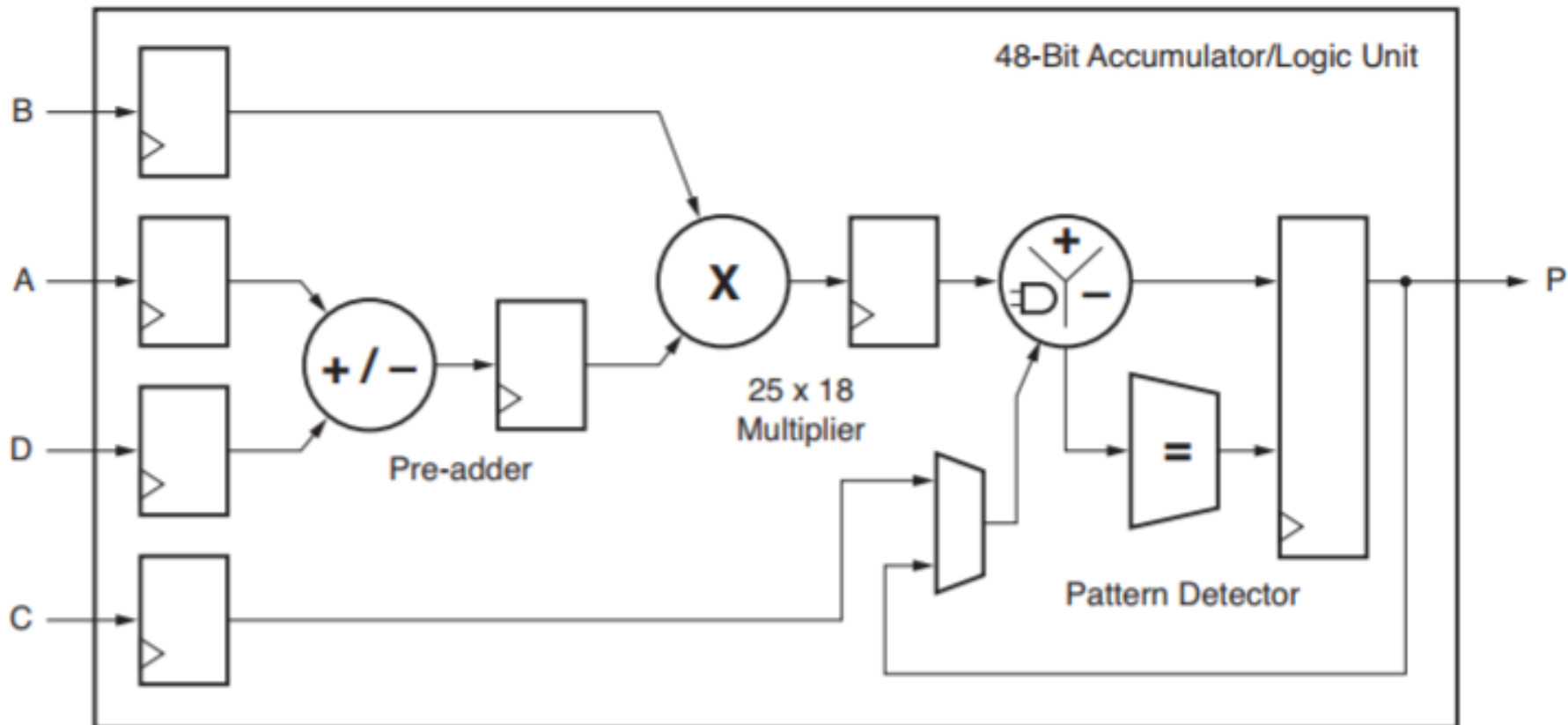
```
always @(posedge <clockA>)
    if (<enableA>)
        if (<write_enableA>)
            <ram_name>[<addressA>] <= <input_dataA>;

always @(posedge <clockB>)
    if (<enableB>)
        <output_dataB> <= <ram_name>[<addressB>];
```


DSP48 Slice

- FPGAs are efficient for DSP applications because they can implement custom, fully parallel algorithms
- DSP applications use many binary multipliers and accumulators that are best implemented in dedicated DSP slices

DSP48 Slice



DSP48 Slice features

- 25×18 two's-complement multiplier
- 48-bit accumulator
 - Can be used as a synchronous up/down counter
- Power saving pre-adder
 - Optimizes symmetrical filter applications
- Dual 24-bit or quad 12-bit add/subtract/accumulate
- Optional logic unit
 - Can generate any one of ten different logic functions of the two operands
- Pattern detector

Q&A